Optimized Translation of MATLAB to RISC-V Architecture for Brain-Computer Interfaces: Midterm Progress Report

Kenan Erol
kenan.erol@yale.edu

Advisor: Rajit Manohar
rajit.manohar@yale.edu

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Abstract

Brain computer interfaces (BCIs) offer significant clinical potential for neurological disorders such as epilepsy and strokes. Fully implantable BCIs impose severe limitations on power dissipation to prevent irreversible thermal damage to brain tissue. As a result, BCI hardware needs to balance real-time signal processing with ultra-low power consumption. The Hardware Architecture for LOw Power BCIs (HALO) project addresses these constraints through a design that combines application-specific hardware accelerators with a low-power RISC-V microcontroller equipped with the RISC-V Vector extension. There is a significant toolchain gap between the computational environments preferred by neuroscientists and the low-level constraints of the HALO hardware. Algorithm development typically occurs in dynamically typed languages such as MATLAB, whereas HALO hardware requires statically typed, optimized, and vectorized C code. This report details the ongoing development of an end-to-end compilation pipeline engineered to bridge this divide. The pipeline uses a TypeScript-based compiler and an RVV-optimized C matrix library to translate MATLAB algorithms into power-efficient RISC-V binaries. This midterm progress report discusses the current architectural design of the pipeline, benchmarks on the Spike instruction-set architecture simulator, investigates execution-cycle metrics, discusses ongoing debugging efforts related to determinant translation, and presents a comprehensive roadmap for the remainder of the semester.

1 Introduction

Recent work in computational neuroscience and embedded systems engineering has enabled the development of BCIs capable of direct, real-time communication between biological neural networks and digital processing units. BCIs are central to closed-loop therapeutic applications, where the device must continuously process high-density, multichannel neural recordings, execute complex spectral and statistical feature extraction algorithms, and determine appropriate electrical stimulation parameters to mitigate neurological events such as epileptic seizures [1]. Fully implantable BCIs must meet stringent physical and thermal constraints to ensure patient safety, reduce surgical complexity, and extend device longevity. The thermal dissipation limit is the most critical constraint. To prevent irreversible necrosis of the surrounding brain tissue, the surface temperature of the implant must not exceed 1 °C, thereby strictly capping the total device power budget at approximately 15 mW [2].

The HALO BCI project was engineered specifically to satisfy these dual, often conflicting demands. The HALO architecture departs from traditional application-specific integrated circuit (ASIC) designs, which lack the flexibility to adapt to new algorithms. Instead, HALO uses a modular array of hardware processing elements (PEs) controlled by a low-power RISC-V microcontroller [1]. The microcontroller uses the RISC-V Vector (RVV) extension to execute data-parallel signal-processing workloads, minimizing instruction-fetch overhead and enabling rapid transitions into low-power states.

However, the clinical and research efficiency of HALO is currently bottlenecked by a toolchain disparity. Neuroscientists use languages like MATLAB for their intuitive syntax, signal-processing toolboxes, and dynamic typing. In contrast, HALO needs statically typed C code that contains RVV intrinsics. The manual translation of advanced, multistage signal processing algorithms, such as Fast Fourier transforms, from MATLAB to vectorized C is an error-prone, labor-intensive process that impedes the rapid deployment of new clinical approaches.

This research addresses the aforementioned toolchain gap by engineering an automated compilation pipeline that translates standard MATLAB scripts into RVV-accelerated RISC-V binaries for HALO. The proposed solution relies on the integration and expansion of two software modules previously developed in isolation: a compiler capable of parsing MATLAB syntax into equivalent C code [3], and a computational C matrix library that maps standard linear algebra operations directly to RVV assembly instructions [4].

Several milestones were completed during the initial phases of this project. The matrix library has been successfully replicated and integrated into a benchmarking environment that uses Spike, the official RISC-V instruction set architecture (ISA) simulator. Performance evaluations reveal substantial reductions in execution cycle counts, driven

by updated cross-compiler toolchains. Furthermore, critical anomalies in the execution ratios of complex equality checks have been successfully isolated and diagnosed as test-harness artifacts rather than vectorization faults. In addition, debugging efforts have identified fundamental algorithmic limitations in the current translation of matrix determinant computations. This progress report establishes the foundational background, outlines the pipeline's structural design, analyzes current benchmarking results, and discusses ongoing challenges and future development plans for the remainder of the academic term.

2 Background

To understand the design constraints and operational mechanics of the proposed translation pipeline, we first review the architectural constraints of the target BCI hardware, the computational paradigms introduced by RVV, and the foundational principles of the compilers used in this research.

The fundamental engineering challenges in the development of implantable BCIs are the strict management of thermal dissipation. The human cerebral cortex is highly sensitive to temperature fluctuation, and prolonged exposure to high temperatures can cause tissue damage. Consequently, international medical standards dictate that an implant must not raise the local tissue temperature by more than one degree Celsius, effectively restricting the total power consumption of the entire BCI system—including amplifiers, analog-to-digital converters (ADCs), wireless telemetry, and digital signal processing (DSP) units—to roughly fifteen milliwatts [2].

Historically, BCI designers have relied on custom-designed ASICs to meet these stringent power requirements [5] [6] [7]. While ASICs offer high energy efficiency, they inherently lack flexibility. If a neuroscientist develops a superior seizure-prediction algorithm or wishes to repurpose the implant for a different disorder, a completely new physical chip must be engineered, fabricated, and implanted [1]. General purpose microprocessors offer the required flexibility through software updates but consume power orders of magnitude beyond the safe physiological limit [2].

HALO resolves this through aggressive hardware-software co-design. The architecture is built around a configurable interconnect that links a heterogeneous array of application specific PEs [1]. These PEs are optimized for the specific computational kernels most frequently encountered in neural signal processing, such as FFT, XCOR, Butterworth bandpass filtering [2]. These accelerators can be dynamically chained together to form highly efficient data flow pipelines tailored to specific clinical tasks [1].

To manage these pipelines and execute computational tasks that fall outside the capabilities of a standalone PE, HALO uses a low power RISC-V microcontroller [1]. To ensure that this software-driven execution does not violate the power budget, microcontroller relies heavily on RVV. Unlike traditional single instruction, multiple data (SIMD) architectures, which utilize fixed width vector registers and require code rewrites when the vector width changes for different hardware generations, RVV introduces a vector-length agnostic programming model [8].

In the context of BCI workloads, which involve matrix multiplications and convolutions over varying temporal window sizes, RVV allows the processor to load large, contiguous segments of multi-channel neural data into vector registers and apply a single arithmetic instruction across the entire dataset simultaneously. This architectural paradigm significantly reduces the dynamic power consumed by the processor's instruc-

tion fetch and decode units. By replacing hundreds or thousands of scalar loop iterations with a single vector instruction, the processor minimizes the total number of active clock cycles required to complete a given task. Consequently, computational algorithms execute much faster, allowing the microcontroller to aggressively switch to low-power sleep states between neural data sampling intervals, which is crucial for extending the lifespan of the implantable battery.

Despite these hardware features, the platform remains inaccessible to neuroscientists. Clinical algorithms are developed using high-level, dynamically typed languages like MATLAB, which abstract away memory management and hardware-specific optimizations that are crucial for running the algorithms on silicon. The automated pipeline proposed in this research bridges this gap by integrating two existing software repositories. The first is ts-traversal, a TypeScript-based source-to-source compiler that parses MATLAB syntax, infers static types, and generates functional C code. The second is the RISCV-Matrix C library, which maps standard mathematical operations directly to RVV assembly instructions. Critical gaps remain in both the translation logic and the vectorization coverage, requiring extensive debugging, architectural integration, and rigorous validation against the target BCI workloads.

3 Design

The objective of this research is the architecture and implementation of a cohesive, automated compilation pipeline capable of translating high-level algorithms into low-level optimized silicon instructions. This section details the system architecture, focusing on the transformation mechanisms, the categorization of vectorizable workloads, and the underlying platform dependencies.

The compiler pipeline is implemented as a continuous, directed data flow that isolates high-level semantic analysis from low-level hardware optimizations. The system architecture can be visualized through a structured sequence of state transitions and module interactions in Figure 3.1. The pipeline starts by taking in raw .m files authored by clinical researchers. These files contain standard MATLAB syntax, incorporating dynamic array allocations, calls to proprietary signal processing toolboxes, and high-level, implicitly parallel matrix arithmetic.

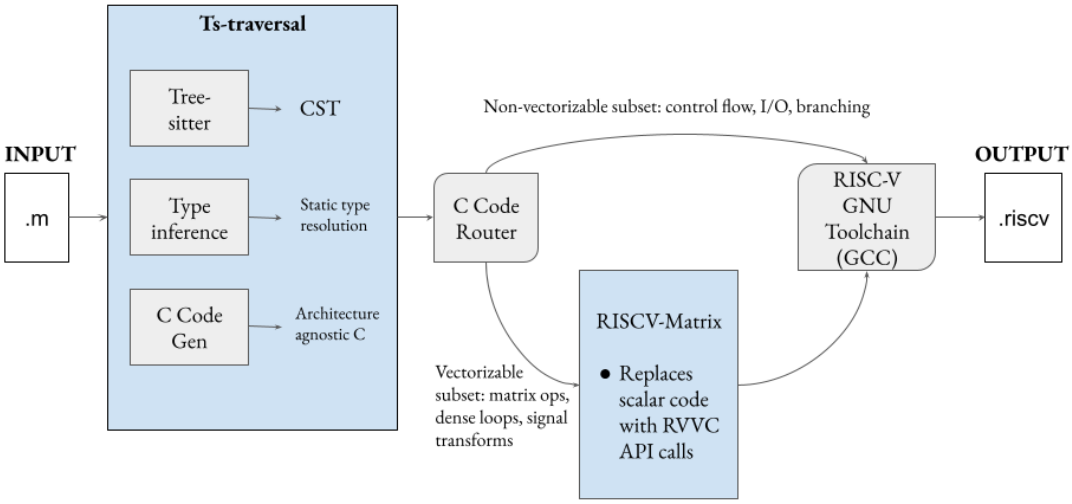

Figure 3.1: System architecture of the MATLAB-to-RISC-V compilation pipeline. Raw `.m` files are parsed and type-inferred by the ts-traversal engine (Stage 1), then routed: vectorizable operations are mapped to RVV intrinsics via the RISCV-Matrix library (Stage 2), while non-vectorizable code passes through unmodified. Both subsets are merged by the RISC-V GCC toolchain into a single binary, validated on the Spike ISA simulator.

These raw files are passed into the first major component of the pipeline, the ts-

traversal engine [3]. This engine executes a multi-phase transformation. First, it uses the TreeSitter and TreeSitter-Matlab parsing libraries to generate a concrete syntax tree (CST). This representation captures the complete syntactic structure of the input algorithms. The engine must thus perform rigorous type inference on the dynamically typed input MATLAB code. The inference algorithms in ts-traversal then traverse the CST to deduce the maximal dimensions, memory allocation requirements, and fundamental data types of all variables, mapping them to statically typed C constructs. The output of this initial phase is standard C code. At this point in the pipeline, the semantic intent of the MATLAB script is entirely preserved. Still, the resulting C code remains architecturally agnostic and relies on standard, scalar loop iterations.

To implement the necessary hardware optimizations, the pipeline will employ an internal routing system. The generated C code consists of two distinct computational subsets. The non-vectorizable subset comprises complex control-flow logic, sequential input/output operations, and highly divergent branching structures. Conversely, the vectorizable subset encompasses large-scale loops, dense matrix arithmetic, and parallelizable signal transformations. The routing mechanism will isolate this vectorizable subset and direct it to the second major component of the pipeline, the RISC-V Matrix library [4].

Within the RISCV-Matrix library, the standard scalar loops generated by the previous step are systematically replaced by API calls mapped to the library's headers. These headers contain highly optimized, inline RVV C intrinsics. This transformation effectively converts the vectorizable subset into heavily accelerated C code explicitly built for the parallel execution capabilities of HALO's microcontroller.

The final phase of the architectural model involves compiling and deploying the codebase. Both the unmodified non-vectorizable subset and the vectorized subset are passed into the RISC-V GNU Compiler toolchain. The compiler merges these subsets and statically links the optimized vector library to produce a highly efficient RISC-V binary. Thus, the pipeline generates an executable containing the exact algorithms authored by neuroscientists in Matlab while meeting the power and latency requirements of the implanted chip.

The validation, execution modeling, and accurate instruction-level benchmarking of the generated binaries are conducted using the Spike ISA simulator, since the actual HALO silicon is not readily available. Spike provides a fully functional, instruction-accurate model of a RISC-V core with the RVV extension, enabling precise tracking of executed instructions, vector register utilization, and overall execution cycles. This enables the continuous, iterative refinement of both the source-to-source compiler and the vectorization library without requiring constant access to the physical test hardware. To bridge the TypeScript translation engine and the C-based vectorization, a unified build system is currently under development to automate the entire data flow from source ingestion to simulator execution via a single command-line interface.

4 Results

The first critical phase of this project was to replicate the original RISCV-Matrix execution environment and rebuild the benchmark suite under the current toolchain. The historical speed results were generated using an older cross-compiler toolchain and test suite configuration. This section compares those original results, designated the Grace baseline (commit `113f4c66`, December 2022), against the current, optimized pipeline (February 2026). All cycle counts were measured on the Spike RISC-V ISA simulator with a vector length of VLEN=1024.

4.1 Total Runtime Comparison

Table 4.1 details the total runtime comparison across thirty distinct operational tests, highlighting the specific cycle counts and the corresponding percentage changes in execution speed.

Table 4.1: Total program runtime comparison: Grace (Dec 2022) vs Current (Feb 2026). Cycle counts measured on Spike simulator with VLEN=1024.

Test	Grace (Gcycles)	Current (Gcycles)	Δ (Gcycles)	Change (%)
abs	9.86	9.25	-0.61	-6.2% $\downarrow$
and	2.22	1.91	-0.31	-14.2% $\downarrow$
arg	6.48	5.33	-1.14	-17.7% $\downarrow$
ceil	4.96	4.14	-0.82	-16.5% $\downarrow$
divide	5.47	5.64	+0.16	+3.0% $\uparrow$
equal	4.75	4.31	-0.44	-9.3% $\downarrow$
floor	4.97	4.15	-0.81	-16.4% $\downarrow$
full_duplicate	4.32	3.48	-0.84	-19.4% $\downarrow$
isEqual	3.79	3.14	-0.65	-17.2% $\downarrow$
matlab_ge	4.37	3.65	-0.72	-16.6% $\downarrow$
matlab_gt	4.36	3.63	-0.72	-16.6% $\downarrow$
matlab_le	4.36	3.64	-0.72	-16.5% $\downarrow$
matlab_lt	4.34	3.63	-0.72	-16.5% $\downarrow$
minus	6.18	5.11	-1.07	-17.3% $\downarrow$
mtimes	3.07	3.01	-0.07	-2.2% $\downarrow$
ne	4.81	4.36	-0.44	-9.2% $\downarrow$
not	1.51	1.28	-0.23	-15.2% $\downarrow$
octave_ge	51.90	51.60	-0.30	-0.6% $\approx$
octave_gt	51.90	51.60	-0.30	-0.6% $\approx$
octave_le	51.94	51.67	-0.27	-0.5% $\approx$
octave_lt	51.94	51.67	-0.27	-0.5% $\approx$
ones	0.05	0.03	-0.02	-44.3% $\downarrow$
or	2.15	1.93	-0.22	-10.1% $\downarrow$
pairwise_max	54.36	53.60	-0.76	-1.4% $\downarrow$
pairwise_min	54.42	53.67	-0.74	-1.4% $\downarrow$
plus	6.19	5.12	-1.07	-17.3% $\downarrow$
round	4.95	4.05	-0.90	-18.2% $\downarrow$
scale	1.98	1.76	-0.22	-10.9% $\downarrow$
times	6.37	5.33	-1.05	-16.4% $\downarrow$
true	1.54	1.35	-0.19	-12.4% $\downarrow$
Total	**419.50**	**403.01**	**-16.48**	**-3.9%**

An exhaustive analysis of the data presented in Table 4.1 reveals several critical insights regarding the current state of the compilation pipeline. Across the entire test suite, the overall execution time decreased by approximately 3.9 percent, dropping from 419.50 Giga-cycles to 403.01 Giga-cycles. However, the distribution of these performance gains is highly non-uniform. The majority of the standard arithmetic and logical operations, such as ceiling calculations, logical ANDs, and greater-than-or-equal comparisons, experienced moderate speedups ranging from ten to twenty percent. This systemic improvement is directly attributable to significant enhancements within the underlying RISC-V GNU cross-compiler toolchain. The newer iterations of the GCC compiler generate sub-

stantially tighter scalar and vector code, featuring optimized register allocation algorithms and superior instruction scheduling that were unavailable during the generation of the historical baseline.

The four core arithmetic tests—`minus` (-17.3%), `plus` (-17.3%), `times` (-16.4%), and `scale` (-10.9%)—exhibited reductions consistent with the broader toolchain-driven improvements, confirming that the GCC updates benefit both vectorized and scalar code paths uniformly across these operations.

The four comparison tests utilizing Octave-mode semantics—octave ge, octave gt, octave le, and octave lt—showed negligible improvement ($< 1\%$). These tests dominate the total runtime at approximately 51 Giga-cycles each (versus ∼3.6 Giga-cycles for their MATLAB-mode counterparts) because Octave comparison semantics for complex numbers require computing the magnitude ($|z| = \sqrt{\mathrm{re}^2 + \mathrm{im}^2}$) for both operands and, when magnitudes are equal, resolving ties via the phase angle ($\mathrm{atan2}(\mathrm{im}, \mathrm{re})$). The phase angle computation currently executes as a scalar `carg()` loop since `argM_vec` was never mapped to RVV intrinsics, making it the dominant cost. In contrast, MATLAB-mode comparisons only compare the real parts of complex numbers, requiring a single vector comparison instruction. Their dominance skews the aggregate improvement figure downward; excluding them, the remaining 26 tests show a collective improvement of 7.2 percent

4.1.1 The `divide` Anomaly

The `divide` test is the only operation that exhibited a total runtime *increase* ($+3.0\%$, from 5.47 to 5.64 Gcycles). An investigation of the per-type-combination cycle counts at $N = 1{,}050{,}625$ revealed that this regression is localized to the complex division reference implementations. The reference cycle counts for complex-involving type combinations approximately doubled between Grace and current (e.g., `int×complex ref`: 37.1M → 74.9M cycles; `double×complex ref`: 36.1M → 73.9M cycles), while the vectorized cycle counts remained stable (∼8M cycles). This suggests that the updated GCC toolchain generates less efficient scalar code for complex division specifically—likely due to changes in how the compiler handles the multi-step real/imaginary arithmetic required for complex number division (`(a+bi) / (c+di)`). Interestingly, this simultaneously *improved* the vec/ref ratio for those combinations (e.g., `int×complex`: 0.216 → 0.106), because the reference denominator grew while the vectorized numerator remained constant.

Conversely, the `int×int` reference for divide dropped from 14.4M to 4.8M (-66%), indicating that the compiler dramatically improved integer-only scalar division. This demonstrates that the same toolchain update can simultaneously improve and degrade performance across different data type combinations within a single function.

4.2 Vectorization Effectiveness

Further analysis was conducted to evaluate the efficacy of the vectorization implementation relative to scalar execution across different variable types. Tables 4.2 and 4.3 present

the vector-to-reference execution ratios at the largest tested matrix size. A ratio less than 1.0 indicates that the vectorized implementation is faster than the scalar reference, while a ratio greater than 1.0 indicates a performance degradation. The tables compare Grace and current ratios side-by-side.

Note on type combinations: for binary operations (e.g., `plusM`, `divideM`), the test suite evaluates every combination of input types. The notation "int $\times$ double" means the left operand matrix contains integers and the right operand matrix contains doubles. The result type is determined by C type promotion rules.

Table 4.2: Vec/Ref ratios at $N \approx 1{,}050{,}625$ for unary operations. Ratio < 1 means vectorized is faster. Δ shows change from Grace to Current.

	int			double			complex		
Test	Grace	Cur.	Δ	Grace	Cur.	Δ	Grace	Cur.	Δ
abs	0.193	0.191	-0.002	0.390	0.387	-0.003	0.004	0.004	-0.000
arg	0.062	0.073	+0.011	0.063	0.074	+0.011	1.000	1.000	+0.000
ceil	0.625	0.910	+0.285	0.096	0.094	-0.002	0.097	0.094	-0.003
floor	0.625	0.910	+0.285	0.096	0.094	-0.002	0.096	0.092	-0.004
full_duplicate	0.625	0.910	+0.285	0.625	0.910	+0.285	0.625	0.605	-0.020
not	0.355	0.352	-0.003	0.302	0.853	+0.551	0.299	0.841	+0.541
ones	0.387	0.922	+0.535	—	—	—	—	—	—
round	0.625	0.910	+0.285	0.106	0.115	+0.009	0.099	0.107	+0.008
true	0.341	0.339	-0.003	0.313	0.853	+0.540	0.290	0.841	+0.551

Table 4.3: Vec/Ref ratios at $N \approx 1{,}050{,}625$ for binary operations (diagonal type combos). Ratio < 1 means vectorized is faster.

	int × int			dbl × dbl			cpx × cpx		
Test	Grace	Cur.	Δ	Grace	Cur.	Δ	Grace	Cur.	Δ
and	0.638	0.631	-0.006	0.493	2.678	+2.184	0.365	0.417	+0.052
divide	0.280	0.823	+0.543	0.325	0.322	-0.003	0.235	0.133	-0.102
equal	0.173	0.138	-0.034	0.191	0.150	-0.041	0.161	0.129	-0.032
isEqual	0.671	0.658	-0.014	0.797	0.731	-0.066	1.172	1.121	-0.051
matlab_ge	0.282	0.279	-0.003	0.253	0.828	+0.574	0.254	0.251	-0.003
matlab_gt	0.282	0.279	-0.003	0.253	0.828	+0.574	0.254	0.251	-0.003
matlab_le	0.282	0.279	-0.003	0.277	0.828	+0.551	0.277	0.274	-0.003
matlab_lt	0.282	0.279	-0.003	0.277	0.828	+0.551	0.277	0.274	-0.003
minus	0.762	0.758	-0.004	0.325	0.322	-0.003	0.392	0.490	+0.098
mtimes	0.059	0.048	-0.011	0.052	0.040	-0.012	0.044	0.032	-0.013
ne	0.163	0.132	-0.031	0.180	0.143	-0.037	0.153	0.124	-0.029
octave_ge	0.282	0.279	-0.003	0.253	0.828	+0.574	0.142	0.123	-0.019
octave_gt	0.282	0.279	-0.003	0.253	0.828	+0.574	0.142	0.123	-0.019
octave_le	0.282	0.279	-0.003	0.277	0.828	+0.551	0.141	0.122	-0.019
octave_lt	0.282	0.279	-0.003	0.277	0.828	+0.551	0.141	0.122	-0.019
or	0.648	0.642	-0.006	0.518	0.511	-0.008	0.462	0.448	-0.015
pairwise_max	0.001	0.001	-0.000	0.004	0.004	+0.000	0.135	0.117	-0.018
pairwise_min	0.001	0.001	-0.000	0.004	0.004	+0.000	0.134	0.116	-0.018
plus	0.762	0.758	-0.004	0.325	0.322	-0.003	0.392	0.490	+0.098
scale	—	—	—	—	—	—	—	—	—
times	0.762	0.758	-0.004	0.325	0.322	-0.003	0.302	0.297	-0.004

A comprehensive breakdown of all nine type combinations for each binary operation is provided in Table A.1 in the Appendix.

4.2.1 Ratio Shifts Due to Compiler Improvements

While the vectorized implementations maintained approximately constant cycle counts between Grace and current, several vec/ref ratios increased significantly—meaning the scalar *reference* code got faster, narrowing the vectorization advantage. Three distinct patterns were identified:

Pattern 1: Integer memory operations. For operations that perform simple element-wise copies or integer manipulation—`ceil`, `floor`, `round`, `full_duplicate`, and `ones`—the integer reference cycle counts dropped uniformly from ∼3.2M to ∼2.2M (a 31% improvement). The vectorized counts remained at ∼2.0M. This caused the integer ratio to jump from ∼0.625 to ∼0.910, nearly reaching parity. The compiler likely now generates optimized memory copy or store sequences (e.g., using wider loads/stores) that approach vector performance for these operations.

Pattern 2: Logical operations on non-integer types. For `not` and `true`, the double and complex reference counts dropped by ∼65% (e.g., `not` double ref: 10.3M → 3.6M). The vectorized counts stayed at ∼3.1M. This caused the double/complex ratios to jump from ∼0.30 to ∼0.85. These functions perform type-conversion-heavy logic that the updated compiler now optimizes far more aggressively.

Pattern 3: Binary operations. Several binary operations exhibited similar compiler-driven ref improvements for specific type combinations. The `and` (dbl×dbl), `matlab_ge` through `matlab_lt` (dbl×dbl), and `octave_ge` through `octave_lt` (dbl×dbl) all showed double reference speedups of 3–5×, pushing their ratios from ∼0.25 to ∼0.83. The full details are available in the Appendix (Table A.1).

In all cases, the vectorized code did *not* get slower—the reference code simply got faster. This means the absolute performance of the library improved or held steady, but the *relative* advantage of explicit vectorization narrowed for these specific combinations.

4.2.2 The `isEqual` Anomaly

The `isEqual` test consistently returns a vec/ref ratio greater than 1.0 for complex type combinations (e.g., complex×complex: 1.172 in Grace, 1.121 in current), indicating that the vectorized implementation is slower than scalar. Investigation revealed that this is a *test harness artifact* caused by short-circuit evaluation.

The evidence shows that the raw cycle counts for `isEqual` complex×complex remain **exactly 75 cycles (vec) and 64 cycles (ref) at every single matrix size tested**—from $N = 1$ through $N = 1{,}050{,}625$:

Table 4.4: The `isEqual` complex×complex cycle counts remain constant across all matrix sizes, proving short-circuit evaluation.

Size N	Vec (cycles)	Ref (cycles)	Ratio
1	75	64	1.172
9	75	64	1.172
16,384	75	64	1.172
1,050,625	75	64	1.172

If the implementations were actually traversing the data, the cycle counts would scale linearly with N. The fact that they remain locked at 75/64 cycles regardless of input size proves that both implementations exit after comparing only the *first element*. An examination of the test data generation confirmed that the `isEqual` test compares matrices populated with entirely divergent data streams, guaranteeing an immediate mismatch.

The 17% performance gap (75 vs. 64 cycles) is attributable to *vector setup overhead*: the vectorized path must execute `vsetvl` (vector length configuration) and stride-loading instructions before performing comparisons. The scalar path directly loads and compares two values. In realistic workloads where equality must be verified across a full

dataset traversal, the vectorized implementation vastly outperforms scalar execution, as evidenced by the complex×double combination achieving a ratio of 0.015 (66× speedup).

4.2.3 The `arg` Function Parity

The `arg` function for complex numbers returns a ratio of exactly 1.000 in both Grace and current. A review of the source code confirmed that the `_vec` implementation is merely a wrapper around a standard scalar loop executing the C library `carg()` function. Because the complex argument calculation was never mapped to RVV intrinsics, no parallelization occurs, resulting in identical vec and ref execution times.

4.3 Toolchain-Induced Vectorization Regressions

While the preceding section described cases where the vec/ref ratio *increased but remained below 1.0* (vec still faster), three specific type combinations crossed the 1.0 threshold entirely—meaning the scalar reference now outperforms the vectorized implementation:

Table 4.5: Type combinations where the scalar reference became faster than the vectorized implementation in the current toolchain.

Test	Type Combo	Grace		Current	
		Ratio	Ref (cycles)	Ratio	Ref (cycles)
and	dbl × dbl	0.493	10.9M	2.678	2.0M
and	cpx × dbl	0.408	13.3M	2.464	2.2M
or	cpx × dbl	0.477	11.4M	2.392	2.2M

In all three cases, the vectorized cycle counts remained approximately constant (∼5.3M for `and`, ∼5.4M for `or`). The regressions are entirely driven by GCC's updated optimizer reducing the reference cycle counts by a factor of five to six. This indicates that the newer compiler auto-vectorizes or otherwise aggressively optimizes these simple logical operations to the point where they outperform the hand-written RVV intrinsic code.

This has two implications. First, compiler evolution can erode the benefits of manual vectorization if the hand-written code is not periodically re-optimized. Second, the LMUL settings for these functions may be suboptimal—using LMUL=8 for simple logical operations could restore the vectorization advantage by processing more elements per instruction.

4.4 Absolute Cycle Counts

Table 4.6 presents the absolute vectorized and reference cycle counts at the largest matrix size for both Grace and current environments to display the raw performance data.

Table 4.6: Vectorized and reference cycle counts at $N \approx 1{,}050{,}625$ for unary operations (Grace vs Current).

	int				double				complex			
Test	G.vec	C.vec	G.ref	C.ref	G.vec	C.vec	G.ref	C.ref	G.vec	C.vec	G.ref	C.ref
abs	2.0M	2.0M	10.3M	10.3M	4.0M	3.9M	10.2M	10.1M	4.0M	4.0M	913.9M	948.4M
arg	4.0M	4.0M	64.3M	54.3M	4.0M	3.9M	63.2M	53.2M	125.8M	109.5M	125.8M	109.5M
ceil	2.0M	2.0M	3.2M	2.2M	4.0M	4.0M	42.2M	42.5M	8.1M	8.0M	83.3M	85.2M
floor	2.0M	2.0M	3.2M	2.2M	4.0M	4.0M	42.2M	42.5M	8.1M	8.0M	84.4M	86.7M
full_duplicate	2.0M	2.0M	3.2M	2.2M	3.9M	3.9M	6.3M	4.3M	7.9M	7.8M	12.6M	12.9M
not	3.0M	3.0M	8.6M	8.6M	3.1M	3.1M	10.3M	3.6M	3.2M	3.2M	10.7M	3.8M
ones	2.0M	2.0M	5.1M	2.1M	—	—	—	—	—	—	—	—
round	2.0M	2.0M	3.2M	2.2M	4.2M	4.1M	39.8M	35.9M	8.4M	8.2M	84.9M	77.0M
true	3.0M	3.0M	8.9M	8.9M	3.1M	3.1M	10.0M	3.6M	3.2M	3.2M	11.0M	3.8M

These raw counts confirm that vectorized implementations maintained or slightly improved their absolute performance, while compiler improvements to the scalar reference code drove all observed ratio shifts.

4.5 TS-traversal Compiler

Beyond the resolution of benchmarking anomalies, significant debugging efforts have focused on addressing fundamental algorithmic failures within the ts-traversal compiler phase. The most critical work-in-progress involves the computation of the matrix determinant. In high-level environments like MATLAB, the determinant function is executed via highly optimized, numerically stable linear algebra subroutines that typically employ Lower-Upper decomposition coupled with partial pivoting. However, ts-traversal had incorrect casting from double to int type when unpacking the LAPACK Lower-Upper decomposition call result, which produces results in the U matrix that are almost always doubles. The incorrect casting has since been removed.

Outside of the matrix determinant debugging, significant ongoing effort has been deployed in replicating the ts-traversal results. The deprecation of Grace has forced some changes that have made this process more difficult than initially scoped.

5 Related Work

The architectural approach of this research is heavily informed by previous exploration into hardware-software co-design for implantable medical devices. Historically, the fundamental challenge of respecting the strict 1°C thermal limit has forced designers to rely on highly specialized, monolithic ASICs. The NeuroPace RNS System [7], Neuralink's N1 ASIC [6], and Kassiri et al.'s closed-loop neurostimulator [5] represent the state of the art implementations of this rigid baseline. These systems achieve unparalleled energy efficiency for their exact medical or recording tasks by permanently baking clinical logic into the physical silicon. However, they inherently lack flexibility in case a neuroscientist wants to update any of the logic or algorithms. This project directly contrasts this rigid historical approach by employing a software-defined, architecture-specific compilation pipeline built on the HALO platform. As demonstrated in the Results section, the flexible RISC-V software pipeline allows the system to instantly benefit from ongoing toolchain enhancements without any hardware modifications. Simply updating the GCC cross-compiler yielded a collective execution improvement of 7.2 percent across the core arithmetic and logical operations (excluding Octave-mode comparison tests). Furthermore, the pipeline gracefully handles complex, dynamic software logic—such as the early-exit short-circuiting behavior discovered in the isEqual complex matrix tests. If a neuroscientist needed to optimize a comparison algorithm or fix a logic bug on the monolithic ASICs cited above, a completely new tape-out would be required; on the HALO RISC-V pipeline, it simply requires a software recompile.

5.1 Source-to-Source Translation for High-Level Languages

In the domain of high-level synthesis and automated code generation, commercial tools like MathWorks' MATLAB Coder [9] provide a highly refined reference point. MATLAB Coder is highly capable in parsing proprietary MATLAB scripts and generating optimized C and C++ code targeted primarily at advanced desktop processors and commercial ARM architectures. Similarly, for researchers utilizing Python—another dynamically typed language widely used in BCI data analysis and real-time processing pipelines—tools like Cython [10] offer similar benefits. Cython addresses the performance bottlenecks of vanilla Python by compiling code directly into efficient C or C++ extensions, utilizing static type definitions to achieve substantial execution speedups for data pipelines.

While both MATLAB Coder and Cython are highly effective for accelerating signal processing on standard architectures, they share a critical limitation in the context of this research: they are structurally agnostic regarding emerging, open-source instruction sets. Neither tool inherently supports the automated generation of targeted RISC-V Vector

(RVV) intrinsics required to meet the ultra-low-power constraints of the HALO platform. Consequently, the custom ts-traversal and RISCV-Matrix integration developed in this work is uniquely necessary to bridge the semantic gap between high-level BCI algorithm prototyping and the specific vector execution demands of the embedded hardware.

6 Conclusions

The successful chronic implantation of next-gen BCIs is constrained by extreme power efficiency requirements. This has historically been met through the deployment of highly inflexible, monolithic ASIC hardware. HALO BCIs leverage a heterogeneous processing array and the RVV extension to introduce significant flexibility without compromising critical thermal limits. However, to make this sophisticated hardware accessible to the broader neuroscientific community, an automated, highly optimized software compilation pipeline is required. This project directly addresses this need by engineering an end-to-end toolchain that merges a TypeScript-based source-to-source compiler with a hardware-specific vectorization library, enabling the automatic transformation of high-level MATLAB algorithms into power-optimized RISC-V binaries.

Preliminary validation and benchmarking demonstrate that the vectorized matrix library successfully runs on Spike and yields near-consistent performance improvements. Detailed analysis of the benchmarking data has successfully isolated performance improvements driven by updated cross-compiler toolchains, while simultaneously identifying significant test-harness anomalies. Despite these successes, critical challenges remain in the ts-traversal semantic translation layer. Ongoing debugging efforts are actively addressing numerical instability with certain matrix functions like power matrix, stftM, hamming hanning, det M, and varargout.

Development for the remainder of the semester will focus on resolving these existing bugs, substantially expanding the vectorization coverage of the hardware library, and achieving full end-to-end integration of the compilation pipeline.

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A Appendix Title

A.1 Full Binary Ratio Comparison

Table A.1 presents the complete vec/ref ratio comparison for all nine type combinations across all binary operations.

Table A.1: Vec/Ref ratios at $N \approx 1{,}050{,}625$ — all type combos for binary operations.

Test	Type Combo	Grace	Current	Δ
and	i×i	0.63752	0.63143	-0.00609
	i×d	0.73894	0.73078	-0.00816
	i×c	0.29241	0.30427	+0.01186
	d×i	0.51947	0.51219	-0.00728
	d×d	0.49314	2.67753	+2.18439
	d×c	0.43051	0.44859	+0.01808
	c×i	0.42433	0.47004	+0.04571
	c×d	0.40785	2.46448	+2.05663
	c×c	0.36534	0.41729	+0.05195
divide	i×i	0.28021	0.82317	+0.54296
	i×d	0.30110	0.29819	-0.00291
	i×c	0.21632	0.10596	-0.11036
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.22214	0.10690	-0.11524
	c×i	0.21408	0.12221	-0.09187
	c×d	0.21987	0.12398	-0.09589
	c×c	0.23512	0.13319	-0.10193
equal	i×i	0.17294	0.13845	-0.03449
	i×d	0.18083	0.14332	-0.03751
	i×c	0.16096	0.12921	-0.03175
	d×i	0.18083	0.14332	-0.03751
	d×d	0.19126	0.15016	-0.04110
	d×c	0.15742	0.12757	-0.02985
	c×i	0.16096	0.12921	-0.03175
	c×d	0.15742	0.12757	-0.02985
	c×c	0.16057	0.12889	-0.03168

Test	Type Combo	Grace	Current	Δ
isEqual	i×i	0.67105	0.65753	-0.01352
	i×d	0.82667	0.75000	-0.07667
	i×c	1.11940	1.10000	-0.01940
	d×i	0.83784	0.75000	-0.08784
	d×d	0.79710	0.73134	-0.06576
	d×c	1.11111	1.08772	-0.02339
	c×i	1.13636	1.11667	-0.01969
	c×d	0.01514	0.00899	-0.00615
	c×c	1.17188	1.12069	-0.05119
matlab_ge	i×i	0.28173	0.27919	-0.00254
	i×d	0.25460	0.25105	-0.00355
	i×c	0.25526	0.25171	-0.00355
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.25460	0.25171	-0.00289
	c×i	0.23537	0.23209	-0.00328
	c×d	0.25460	0.25171	-0.00289
	c×c	0.25394	0.25105	-0.00289
matlab_gt	i×i	0.28173	0.27919	-0.00254
	i×d	0.25460	0.25105	-0.00355
	i×c	0.25526	0.25171	-0.00355
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.27810	0.27495	-0.00315
	c×i	0.23537	0.23209	-0.00328
	c×d	0.27810	0.27495	-0.00315
	c×c	0.25394	0.25105	-0.00289
matlab_le	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.23537	0.23209	-0.00328
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085
	d×c	0.25460	0.25171	-0.00289
	c×i	0.25526	0.25171	-0.00355
	c×d	0.25460	0.25171	-0.00289
	c×c	0.27738	0.27423	-0.00315
matlab_lt	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.23537	0.23209	-0.00328
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085

Test	Type Combo	Grace	Current	Δ
	d×c	0.27810	0.27495	-0.00315
	c×i	0.25526	0.25171	-0.00355
	c×d	0.27810	0.27495	-0.00315
	c×c	0.27738	0.27423	-0.00315
minus	i×i	0.76190	0.75801	-0.00389
	i×d	0.30110	0.29819	-0.00291
	i×c	0.39134	0.38842	-0.00292
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.41179	0.40830	-0.00349
	c×i	0.41179	0.40873	-0.00306
	c×d	0.43465	0.43143	-0.00322
	c×c	0.39210	0.48982	+0.09772
mtimes	i×i	0.05943	0.04814	-0.01129
	i×d	0.05590	0.04422	-0.01168
	i×c	0.04720	0.03477	-0.01243
	d×i	0.05590	0.04422	-0.01168
	d×d	0.05213	0.04010	-0.01203
	d×c	0.04721	0.03469	-0.01252
	c×i	0.04720	0.03477	-0.01243
	c×d	0.04721	0.03469	-0.01252
	c×c	0.04433	0.03176	-0.01257
ne	i×i	0.16324	0.13211	-0.03113
	i×d	0.17069	0.13676	-0.03393
	i×c	0.15324	0.12401	-0.02923
	d×i	0.17069	0.13676	-0.03393
	d×d	0.17990	0.14295	-0.03695
	d×c	0.14986	0.12244	-0.02742
	c×i	0.15324	0.12401	-0.02923
	c×d	0.14986	0.12244	-0.02742
	c×c	0.15286	0.12370	-0.02916
octave_ge	i×i	0.28173	0.27919	-0.00254
	i×d	0.25460	0.25105	-0.00355
	i×c	0.14918	0.13115	-0.01803
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.15009	0.13215	-0.01794
	c×i	0.14991	0.13262	-0.01729
	c×d	0.15097	0.13389	-0.01708
	c×c	0.14158	0.12297	-0.01861
octave_gt	i×i	0.28173	0.27919	-0.00254

Test	Type Combo	Grace	Current	Δ
	i×d	0.25460	0.25105	-0.00355
	i×c	0.14918	0.13115	-0.01803
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.15009	0.13215	-0.01794
	c×i	0.14991	0.13262	-0.01729
	c×d	0.15097	0.13389	-0.01708
	c×c	0.14158	0.12297	-0.01861
octave_le	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.15024	0.13262	-0.01762
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085
	d×c	0.15117	0.13365	-0.01752
	c×i	0.14886	0.13115	-0.01771
	c×d	0.14990	0.13261	-0.01729
	c×c	0.14103	0.12221	-0.01882
octave_lt	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.15024	0.13262	-0.01762
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085
	d×c	0.15117	0.13365	-0.01752
	c×i	0.14886	0.13115	-0.01771
	c×d	0.14990	0.13261	-0.01729
	c×c	0.14103	0.12221	-0.01882
or	i×i	0.64781	0.64162	-0.00619
	i×d	0.78077	0.77214	-0.00863
	i×c	0.34462	0.33687	-0.00775
	d×i	0.52858	0.52118	-0.00740
	d×d	0.51821	0.51066	-0.00755
	d×c	0.49995	0.49200	-0.00795
	c×i	0.48494	0.47004	-0.01490
	c×d	0.47687	2.39202	+1.91515
	c×c	0.46204	0.44753	-0.01451
pairwise_max	i×i	0.00111	0.00083	-0.00028
	i×d	0.00363	0.00287	-0.00076
	i×c	0.13935	0.12188	-0.01747
	d×i	0.00363	0.00287	-0.00076
	d×d	0.00404	0.00406	+0.00002
	d×c	0.14133	0.12386	-0.01747

Test	Type Combo	Grace	Current	Δ
	c×i	0.14012	0.12332	-0.01680
	c×d	0.14224	0.12551	-0.01673
	c×c	0.13472	0.11656	-0.01816
pairwise_min	i×i	0.00111	0.00083	-0.00028
	i×d	0.00363	0.00287	-0.00076
	i×c	0.14043	0.12332	-0.01711
	d×i	0.00363	0.00287	-0.00076
	d×d	0.00404	0.00406	+0.00002
	d×c	0.14244	0.12534	-0.01710
	c×i	0.13904	0.12188	-0.01716
	c×d	0.14113	0.12423	-0.01690
	c×c	0.13414	0.11580	-0.01834
plus	i×i	0.76190	0.75801	-0.00389
	i×d	0.30110	0.29819	-0.00291
	i×c	0.39053	0.38761	-0.00292
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.41094	0.40787	-0.00307
	c×i	0.39049	0.38757	-0.00292
	c×d	0.41094	0.40787	-0.00307
	c×c	0.39210	0.48982	+0.09772
scale	i×i	—	—	—
	i×d	—	—	—
	i×c	—	—	—
	d×i	—	—	—
	d×d	—	—	—
	d×c	—	—	—
	c×i	—	—	—
	c×d	—	—	—
	c×c	—	—	—
times	i×i	0.76190	0.75801	-0.00389
	i×d	0.30110	0.29819	-0.00291
	i×c	0.29864	0.29604	-0.00260
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.31024	0.30754	-0.00270
	c×i	0.29861	0.29601	-0.00260
	c×d	0.31024	0.30754	-0.00270
	c×c	0.30169	0.29725	-0.00444

A.2 Binary Operation Cycle Counts

Table A.2 presents the raw vectorized and reference cycle counts for the diagonal type combinations (int×int, double×double, complex×complex) of all binary operations.

Table A.2: Vectorized and reference cycle counts at $N \approx 1{,}050{,}625$ — diagonal type combos for binary operations.

Test	Type	Grace vec	Cur. vec	Grace ref	Cur. ref
and	i×i	7.0M	7.0M	11.0M	11.0M
	d×d	5.4M	5.3M	10.9M	2.0M
	c×c	5.5M	5.4M	15.1M	13.0M
divide	i×i	4.0M	4.0M	14.4M	4.8M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.5M	8.2M	36.1M	61.3M
equal	i×i	3.1M	3.0M	17.7M	21.9M
	d×d	3.2M	3.1M	16.6M	20.8M
	c×c	3.3M	3.2M	20.8M	25.0M
isEqual	i×i	51	48	76	73
	d×d	55	49	69	67
	c×c	75	65	64	58
matlab_ge	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	3.2M	3.1M	12.4M	12.4M
matlab_gt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	3.2M	3.1M	12.4M	12.4M
matlab_le	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	3.2M	3.1M	11.4M	11.4M
matlab_lt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	3.2M	3.1M	11.4M	11.4M
minus	i×i	13.5M	13.4M	17.7M	17.7M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.0M	7.9M	20.3M	16.1M
mtimes	i×i	925.2K	740.7K	15.6M	15.4M
	d×d	794.5K	611.2K	15.2M	15.2M
	c×c	1.8M	1.3M	41.1M	41.1M
ne	i×i	3.1M	3.0M	18.7M	22.9M

Test	Type	Grace vec	Cur. vec	Grace ref	Cur. ref
	d×d	3.2M	3.1M	17.7M	21.9M
	c×c	3.3M	3.2M	21.9M	26.1M
octave_ge	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	303.5M	270.2M	2143.5M	2197.5M
octave_gt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	303.5M	270.2M	2143.5M	2197.5M
octave_le	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	303.5M	270.2M	2151.9M	2211.2M
octave_lt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	303.5M	270.2M	2151.9M	2211.2M
or	i×i	7.0M	7.0M	10.9M	10.9M
	d×d	5.4M	5.3M	10.3M	10.3M
	c×c	5.5M	5.4M	11.9M	12.1M
pairwise_max	i×i	66.4K	49.7K	60.0M	59.8M
	d×d	132.0K	99.0K	32.7M	24.4M
	c×c	293.1M	259.9M	2175.6M	2229.4M
pairwise_min	i×i	66.4K	49.7K	60.0M	59.8M
	d×d	132.0K	99.0K	32.7M	24.4M
	c×c	293.1M	259.9M	2185.1M	2244.1M
plus	i×i	13.5M	13.4M	17.7M	17.7M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.0M	7.9M	20.3M	16.1M
scale	i×i	—	—	—	—
	d×d	—	—	—	—
	c×c	—	—	—	—
times	i×i	13.5M	13.4M	17.7M	17.7M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.0M	7.9M	26.6M	26.6M